

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

DECEMBER EXAMINATION

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES

FRIDAY, 29 DECEMBER 2023

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates :

This question paper consists of FIVE (5) questions .

Answer **ALL** questions in **Section A** and **ONLY ONE(1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in section B, marks will only be awarded for the FIRST ONE (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Section A (Answer All Questions)**

- Q1. The following table shows the directed distances (km) to travel from a particular town to the other. For cases that two towns are not directly linked, their distance is marked as “-”. For example, the distance from town A to town D is 5km but there is no road going from A to C, E, or F directly, hence denoted as “-”. Please note that $A \rightarrow D$ is one-way in this scenario.

Distance (km)		Town (to)					
		A	B	C	D	E	F
Town (from)	A	0	6	-	5	-	-
	B	6	0	7	-	-	-
	C	-	-	0	-	-	3
	D	-	-	4	0	2	-
	E	-	-	-	-	0	3
	F	-	-	-	-	-	0

- (a) If **Breadth-First Search (BFS)** is used to find the path to go from town A to F,
- Draw the search tree generated by the BFS algorithm. If there are several possible nodes to explore, organize and select the node based on *alphabetical* order from left to right. (5 marks)
 - What is the path and cost found by BFS to go from A to F? (2 marks)
 - Is the path optimal? Justify your answer. (2 marks)
 - Given that the branching factor is approximately 2, what is the time complexity to run BFS? (2 marks)
- (b) If **Uniform Cost Search (UCS)** is used to find the optimal path to go from town A to F,
- Draw the search tree generated by the UCS algorithm. If there exist nodes with the similar cost, organize and select the node based on *alphabetical* order from left to right. (5 marks)
 - What is the path and cost found by UCS to go from A to F? (3 marks)
 - Is the path optimal? Justify your answer. (3 marks)
 - Given that the branching factor is approximately 2, what is the time complexity to run UCS? (3 marks)
- [Total: 25 marks]

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- Q2. (a) Specify the differences between **Supervised** and **Unsupervised Learnings**. (4 marks)
- (b) Table 2.1 shows data on the average yearly income (y) of a person with different number of years (x) of working experience.

Table 2.1

x (year)	y (income, K)
1	25
4	32
7	53
9	75

Suppose that a *linear model* is used to predict y from input x .

- (i) Explain whether this is a *regression* task or a *classification* task. (2 marks)
- (ii) Write down the hypothesis function $h_{\theta}(x)$ using parameters (θ_0, θ_1) for the model. (2 marks)
- (iii) Which of the two models $(\theta_0 = 15, \theta_1 = 6)$ or $(\theta_0 = 30, \theta_1 = 3)$ is a better model for the data in Table 2.1? Justify your answer by computing the **mean square error**. (9 marks)
- (c) Figure 2.1 shows the training error and cross-validation error for a certain dataset when varying the model complexity.

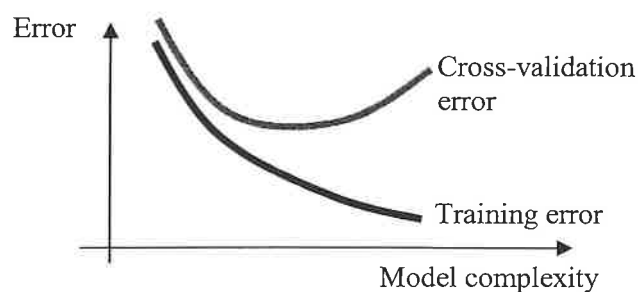


Figure 2.1: Model complexity versus errors.

- (i) Copy Figure 2.1 to your answer sheet and identify the region for **high bias** and **high variance**. (2 marks)
- (ii) Explain how the two phenomena are related to model complexity. (4 marks)
- (iii) If a model overfits the data, suggest TWO approaches to resolve the problem. (2 marks)

[Total: 25 marks]

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- Q3. (a) Table 3.1 shows the predicted scores of a binary classifier versus the actual labels for ten test samples. The threshold value of the classifier is set at 0.5.

Table 3.1: Predicted scores of a binary classifier versus actual labels.

Sample #	Actual Label	Predicted Score
1	1	0.78
2	0	0.55
3	1	0.42
4	1	0.86
5	1	0.92
6	1	0.47
7	0	0.02
8	0	0.34
9	1	0.89
10	1	0.71

Determine the following performance measures for the classifier in Table 3.1. Include your calculation in the answer.

- (i) Accuracy (2 marks)
 - (ii) Precision (2 marks)
 - (iii) Recall (2 marks)
 - (iv) F₁ score (2 marks)
- (b) If you want to improve the *recall* rate of the classifier in (a), should you increase or decrease the threshold value of the classifier? Justify your answer. (2 marks)
- (c) Draw a graphical representation of the Markov Decision Process (MDP) and describe the interactions between the key elements in MDP. (7 marks)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q3. (Continued)**

- (d) Figure 3.1 shows a simple one-dimensional grid world. There are totally five grids (states) in the grid world. The number in each of the grids indicates the reward received if the agent arrives at the grid. The starting state is at grid #2. The terminal state is at grid #5 with a +2 reward. The agent can only move either left (L) or right (R). You are asked to solve this grid world problem using Q-learning algorithm with learning rate $\alpha=0.5$ and discount factor $\gamma=0.9$.

1	2	3	4	5
0	0	0	0	+2

Figure 3.1. One-dimensional grid world.

- (i) Construct the Q-table for the grid world problem. Initialize the Q-table to zeros. (2 marks)
- (ii) In the first training episode, the agent takes [R, L, R, R, R] action sequence to reach the terminal state. Update the Q-table after the first episode, that is for Q(4, R). Show your calculation steps. (2 marks)
- (iii) In the second training episode, the agent takes [R, R, R] action sequence to reach the terminal state. Update the Q-table after the second episode, that is for Q(3, R) and Q(4, R). Show your calculation steps. (4 marks)
- [Total: 25 marks]

Section B (Choose Any ONE (1) Question)

- Q4. (a) Given the following cost equation which we wish to minimize:

$$f(x_1, x_2) = 0.1(x_1 - 2)^2 + 0.2(x_2 - 3)^2$$

- (i) Derive the update equations for x_1 and x_2 using the *gradient descent* formula below, where η denotes the learning rate. (4 marks)

$$x_j := x_j - \eta \frac{\partial}{\partial x_j} f(x_1, x_2) \text{ for } j = 1, 2$$

- (ii) Given initial values $x_1 = 0$, $x_2 = 0$ and learning rate $\eta = 0.5$, compute the cost function f for the first THREE (3) iterations using *gradient descent* algorithm. (6 marks)
- (iii) Verify if gradient descent is behaving properly based on the cost function values in (ii). (1 mark)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q4. (Continued)**

- (b) Table 4.1 shows data on the average lifespan of an engine (denoted as y) when different amount of lubricant (denoted as x) is used during service.

Table 4.1

Lubricant (x)	Engine Lifespan (y)
3	25
5	45
7	53
9	50

Suppose that a **linear** model is used to predict the *Engine Lifespan* based on the amount of *Lubricant* used. Assume *gradient descent* is used to solve the problem. Assuming the initial parameters $\theta_0 = 15$ and $\theta_1 = 2$ and the learning rate $\eta = 0.01$, show the updated values of θ_0 and θ_1 after the **first** iteration. (8 marks)

- (c) Assuming binary classifier is used to perform a multiclass classification task with 10 classes by using the following strategies. Compute the total number of binary classifiers needed for each strategy, and identify ONE advantage and ONE disadvantage for each of the strategies.

(i) One-versus-All (OvA) (3 marks)

(ii) One-versus-One (OvO) (3 marks)

[Total: 25 marks]

- Q5. (a) A school is planning to organize an outdoor event this Sunday. The weather forecast predicted that there is only 20% chance of bad weather on Sunday. From past records, if the weather is bad, only 30% chance that an event will be successfully organized, but if the weather is not bad, 90% chance that the event will be successful. Let S represents that an event is successful and let B represents that the weather is bad.

(i) Compute the following probabilities: $P(S|B)$, $P(S|\neg B)$, $P(\neg S|B)$ and $P(\neg S|\neg B)$. (4 marks)

(ii) What is the probability that the weather is bad and the event is successfully organized? (2 marks)

(iii) What is the probability of the event is successful? (3 marks)

(iv) Given that the event is not successful, what is the probability that the weather is bad? (3 marks)

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUES**Q5. (Continued)**

(b) Figure 5.1 shows a Bayesian Network:

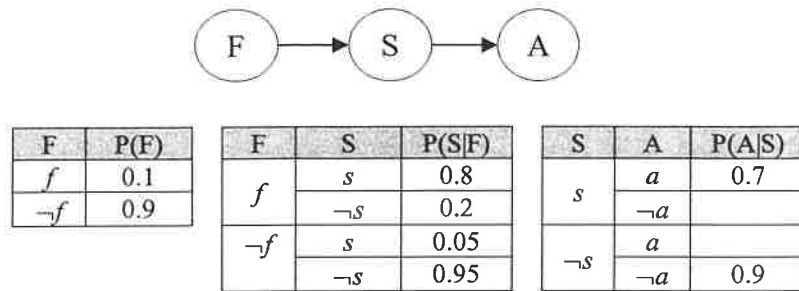


Figure 5.1: Bayesian Network

- (i) Write the equation for the joint distribution $P(F, S, A)$. (3 marks)
- (ii) Identify the conditional independence relationship in the network. (2 marks)
- (iii) Compute the following probabilities: (8 marks)
 - $P(f, s, a)$
 - $P(s, a)$
 - $P(\neg a | f, s)$

[Total: 25 marks]

UCCD2063 ARTIFICIAL INTELLIGENCE TECHNIQUESAppendix

$$Accuracy = \frac{TN + TP}{TN + FP + FN + TP}$$

$$Precision = \frac{TP}{FP + TP}$$

$$F_1 = \frac{TP}{TP + \frac{FN + FP}{2}}$$

$$Recall = \frac{TP}{FN + TP}$$

$$h(\mathbf{x}) = \theta^T \mathbf{x}$$

$$\mathbf{h} = \mathbf{X} \cdot \theta$$

$$MSE(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$RMSE(\theta) = \sqrt{MSE(\theta)}$$

$$\nabla_{\theta} MSE(\theta) = \frac{2}{m} \mathbf{X}^T \cdot (\mathbf{X} \cdot \theta - \mathbf{y})$$

$$h_{\theta}(\mathbf{x}) = \sigma(\theta^T \mathbf{x}) = \frac{1}{1 + e^{-\theta^T \mathbf{x}}}$$

$$\nabla_{\theta} MSE(\theta) = \frac{1}{m} \mathbf{X}^T \cdot (h_{\theta}(\mathbf{x}) - \mathbf{y})$$

$$P(A, B) = P(A|B)P(B)$$

$$P(A, B) = P(B|A)P(A)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A) + P(\neg A) = 1$$

$$Q(s, a) = Q(s, a) + \alpha(R(s, a) + \gamma \max_{a'} Q(s', a') - Q(s, a))$$